

1 Abstract

This project focuses on the design and hardware implementation of a wireless home automation system using the AT89C51 microcontroller and an HC-06 Bluetooth module. The work aims to demonstrate how everyday electrical devices can be controlled remotely via a mobile phone. The implementation uses discrete LEDs to represent home appliances, showcasing how serial communication (UART protocol) translates digital commands from a mobile application into physical hardware actuation.

2 Introduction

With the rapid advancement of the Internet of Things (IoT) and smart home technologies, wireless control of household appliances has become increasingly relevant. This project aims to design a localized, Bluetooth-based control system that bridges the gap between mobile software and embedded hardware. By utilizing an HC-06 Bluetooth module interfacing with the standard 8051 architecture, this project provides a practical realization of serial data reception, command processing, and logical output control.

3 Methodology

The project follows a systematic approach to bridge theoretical embedded systems concepts with practical realization:

1. **System Architecture Design:** Mapping out the connections between the AT89C51, the clock/reset circuits, the HC-06 module, and the output LEDs.
2. **Serial Communication Setup:** Configuring the microcontroller's Timer 1 in auto-reload mode to establish a 9600 bps baud rate for the UART interface.
3. **Control Logic Development:** Writing an Assembly Language program to parse incoming ASCII characters ('0' through '9') and trigger specific I/O pins.
4. **Simulation:** Verifying the logic and serial communication using Proteus Design Suite before hardware assembly.
5. **Hardware Implementation:** Assembling the complete circuit on a breadboard.

4 Tools Used

- **Simulation & Software Tools:** Keil μ Vision (for assembly compilation), Proteus Design Suite (for simulation), and a Mobile Bluetooth Terminal Application.
- **Hardware Components:**
 - AT89C51 Microcontroller
 - HC-06 Bluetooth Module
 - 11.0592 MHz Crystal Oscillator and 30pF capacitors
 - Resistors (10k Ω , 330 Ω) and Capacitor (10 μ F)
 - 4x LEDs (representing home appliances)
 - 5V DC Power Supply and connecting wires

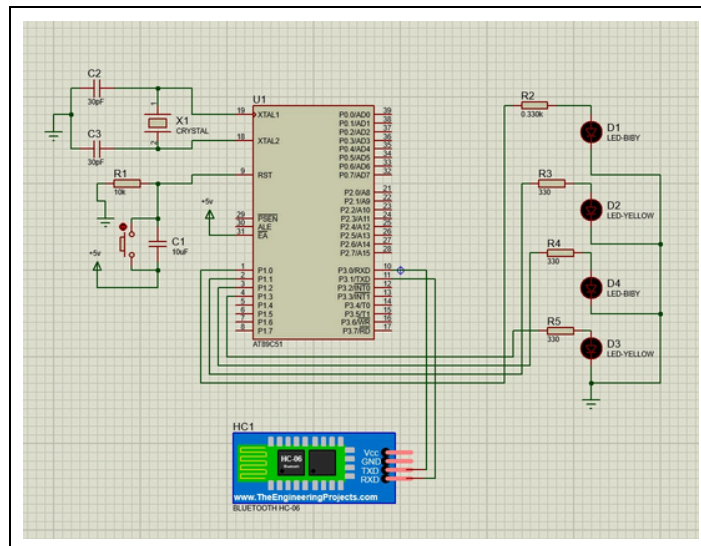


Figure 1: Hardware schematic of the Bluetooth Home Automation System.

5 Result / Work Done

- The circuit was successfully designed, simulated, and implemented in hardware.
- Serial communication between the mobile phone and the HC-06 module was established flawlessly at 9600 baud.
- The system accurately processed the following input commands:
 - Sending '1', '3', '5', '7' individually turned **ON** LEDs 1 through 4.
 - Sending '2', '4', '6', '8' individually turned **OFF** LEDs 1 through 4.
 - Sending '9' successfully turned **ON all** devices simultaneously.
 - Sending '0' successfully turned **OFF all** devices simultaneously.
- The microcontroller effectively acted as a precise electronic switch based on the wireless input.

6 Conclusion

The Bluetooth-based home automation project successfully demonstrated the practical implementation of wireless serial communication using the AT89C51 microcontroller. It provided valuable hands-on experience in configuring UART registers, baud rate generation, and interfacing external communication modules with an embedded processor. This project forms a robust foundation that can easily be scaled up using relays to control actual high-voltage AC appliances in a real smart-home environment.

7 Appendices

Program / Simulation Code

; AT89C51 Assembly Code for Bluetooth Home Automation
ORG 0000H

SJMP MAIN

MAIN:

```
MOV TMOD, #20H      ;Timer1 in mode2(8-bit auto-reload)
MOV TH1, #0FDH       ;Set baud rate to 9600 (11.0592 MHz crystal)
MOV SCON, #50H       ;Serial mode 1, 8-bit data, enable receiver; Start Timer1
SETB TR1             ;Initialize Port1: All devices OFF initially
MOV P1, #00H
```

RECEIVE:

```
JNB RI, RECEIVE      ;Wait here until a character is received
MOVA, SBUF            ;Move received data from SBUF to Accumulator
CLR RI               ;Clear Receive Interrupt flag for next byte
; --- Device 1 Control ---
CJNE A, #'1', S1OFF ; If not '1', jump to check next condition
```

```
SETBP1.0             ;Turn ON Device1
SJMP RECEIVE
```

```
S1OFF: CJNE A, #'2', S2ON
CLR P1.0             ;Turn OFF Device1
SJMP RECEIVE
; --- Device 2 Control ---
```

```
S2ON: CJNE A, #'3', S2OFF
SETBP1.1             ;Turn ON Device2
SJMP RECEIVE
```

```
S2OFF: CJNE A, #'4', S3ON
CLR P1.1             ;Turn OFF Device2
SJMP RECEIVE
; --- Device 3 Control ---
```

```
S3ON: CJNE A, #'5', S3OFF
SETBP1.2             ;Turn ON Device3
SJMP RECEIVE
```

```
S3OFF: CJNE A, #'6', S4ON
CLR P1.2             ;Turn OFF Device3
SJMP RECEIVE
; --- Device 4 Control ---
```

```
S4ON: CJNE A, #'7', S4OFF
SETBP1.3             ;Turn ON Device4
SJMP RECEIVE
```

```
S4OFF: CJNE A, #'8', ALLOFF
CLR P1.3             ;Turn OFF Device4
SJMP RECEIVE
```

```

        ; --- Master Controls ---
ALLOFF: CJNE A,#'0',ALLON
        MOVP1,#00H           ;TurnOFFalldevices
        SJMP RECEIVE
ALLON: CJNE A,#'9',RECEIVE
        MOVP1,#0FH           ;TurnONalldevices(P1.0toP1.3)
        SJMP RECEIVE
        END

```

References

1. Muhammad Ali Mazidi, *The 8051 Microcontroller and Embedded Systems*, Pearson.
2. HC-06 Bluetooth Module Datasheet.
3. AT89C51 Microcontroller Datasheet, Atmel Corporation.